



TULSIRAMJI GAIKWAD-PATIL College of Engineering and Technology

Wardha Road, Nagpur - 441108

Accredited with NAAC A+ Grade

Approved by AICTE, New Delhi, Govt. of Maharashtra



(An Autonomous Institute Affiliated to RTM Nagpur University with NAAC A+ Grade)

Department of Information Technology

Session 2025-26(Even Semester)

VISION	MISSION
To emerge as a learning hub and center of excellence in the domain of Information Technology	[M1] To impart quality technical education through effective teaching learning process. [M2] To provide a platform to address societal issues as well as challenges faced by IT industries. [M3] To foster a culture of research and impart innovative and entrepreneurial skills in the field of IT. [M4] To ensure overall development of students and staff by inculcating knowledge and professional ethics as a part of lifelong learning.

Class Test (CT)-2

Semester: VI

Programme: B. Tech Information Technology

Course: Artificial Intelligence and Machine Learning

Time: 1 Hr 30 min

Max Marks: 30 Marks

Instructions to candidates:

1. All questions are compulsory in Group A
2. Solve any two sub-question out of three in Group B
3. Assume suitable data wherever necessary
4. All question carries marks as indicated

BIT33601	Course Outcome
CO1	Identify knowledge representation, problem solving and learning methods of AI.
CO2	Discuss knowledge and reasoning and its representation in AI.
CO3	Describe artificial intelligence, machine learning and deep learning.
CO4	Analyze supervised learning algorithms and decision trees.
CO5	Evaluate machine learning models using cross-validation and sampling techniques like bootstrapping

Group A				
Que. 1	Answer the following Question	CO	BTL	Marks
a.	Describe sub-field of Machine Learning.	CO3	2	2
b.	Describe data mining in terms of Machine Learning.	CO4	2	2
c.	Interpret Principle Component Analysis in detail.	CO5	2	2

Group B				
Que. 2	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Examine classification problem in detail.	CO3	4	4

b.	Illustrate the basic principle of reinforcement learning.	CO3	2	4
c.	Explain primary goal of regression problem.	CO3	2	4
Que 3.	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Summarize RNNs handle sequential data compared to traditional neural networks.	CO4	2	4
b.	Apply the KNN algorithm to a dataset and evaluate its performance.	CO4	3	4
c.	Describe classification algorithm with suitable example.	CO4	2	4
Que 4.	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Discuss linear discriminant analysis in detail.	CO5	3	4
b.	Outline the use of k-fold cross validation in Machine Learning.	CO5	2	4
c.	Distinguish between K-means and Fuzzy-means algorithm.	CO5	4	4

Bloom's Taxonomy Level (BTL)- Remember , Understand , Apply, Analyze , Evaluate , Create